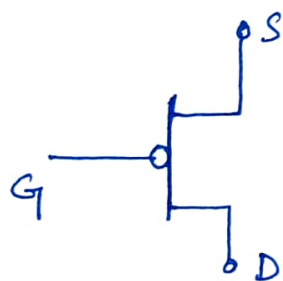


CMOS (Complimentary Metal Oxide Semiconductor) :-

* CMOS As Inverter / NOT Gate :-

⇒ In CMOS As inverter, NMOS & PMOS are Connected in Complimentary fashion.

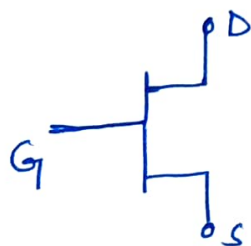
PMOS (p-channel Metal oxide semiconductor)



If $G=0$, S to D $\rightarrow$ short Circuit $\rightarrow$ ON

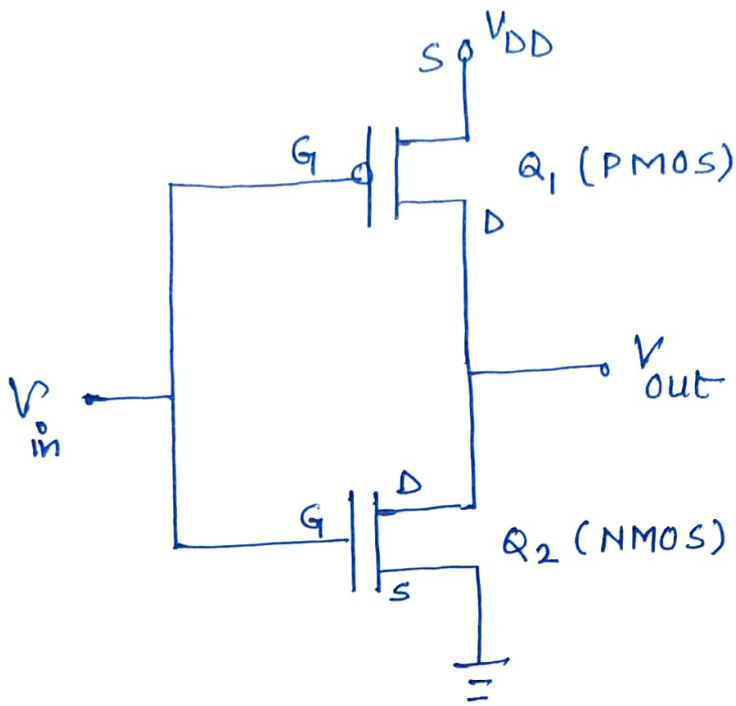
If $G=1$, S to D $\rightarrow$ open Circuit $\rightarrow$ OFF

NMOS (n-channel Metal oxide semiconductor)



If $G=0$, S to D $\rightarrow$ open circuit $\rightarrow$ OFF

If $G=1$, S to D $\rightarrow$ short Circuit $\rightarrow$ ON



Input V_{in}	PMOS	NMOS	V_{out}
0	$\delta \cdot C$	O.C	$V_{DD} (1)$
1	O.C	$\delta \cdot C$	GND (0)

Advantages :-

- ① Circuit is simple
- ② Power usage is less
- ③ Work on 15V to 20V
- ④ Low power dissipation

Disadvantages :-

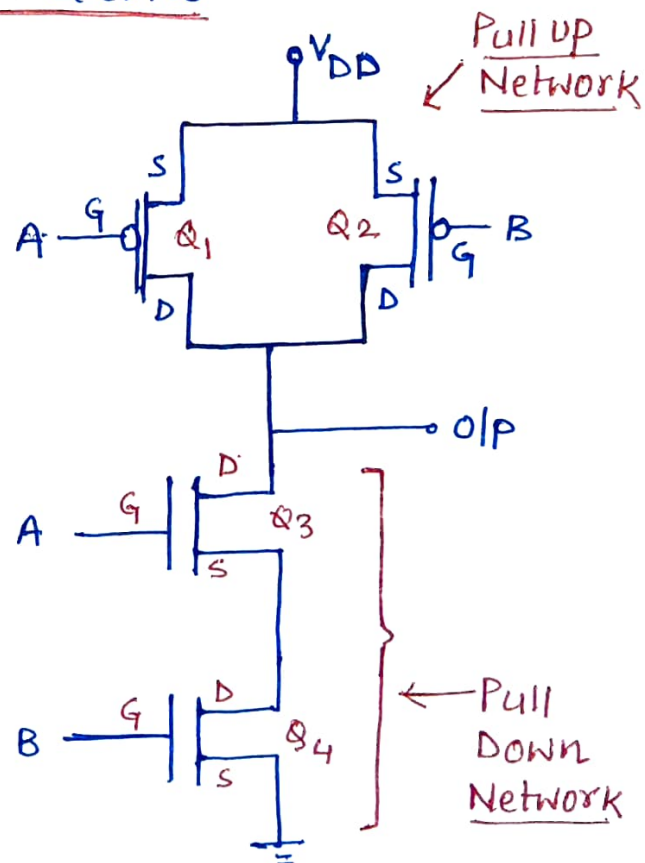
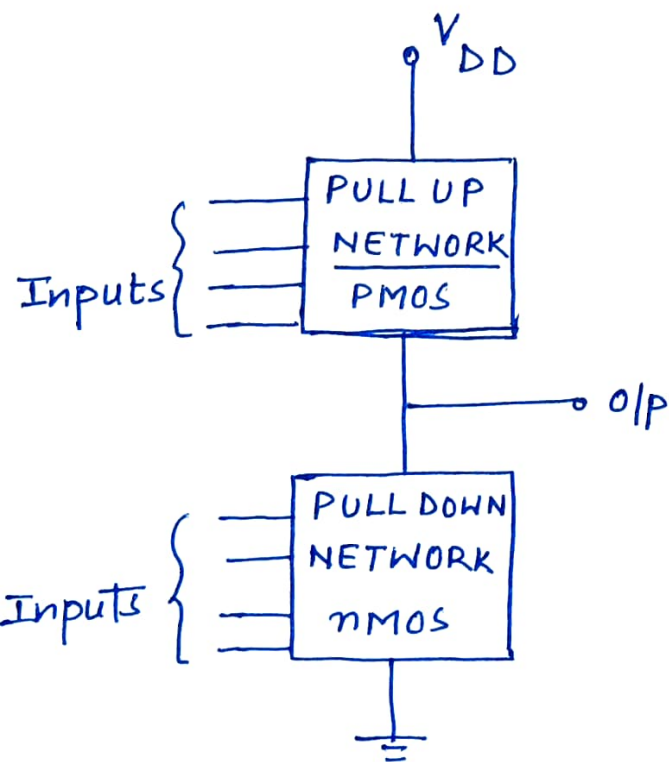
- ① propagation time is more
- ② stored in special conducting foam.

CMOS AS NAND Gate :- NAND Gate :- A B $y = \overline{A \cdot B}$

Truth Table of NAND Gate

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

* In CMOS AS NAND Gate, We have (.) operation therefore PMOS should be connected in parallel & nmos should be connected in series.



Truth Table

(4)

Inputs		PMOS		NMOS		Output
		Q ₁	Q ₂	Q ₃	Q ₄	
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

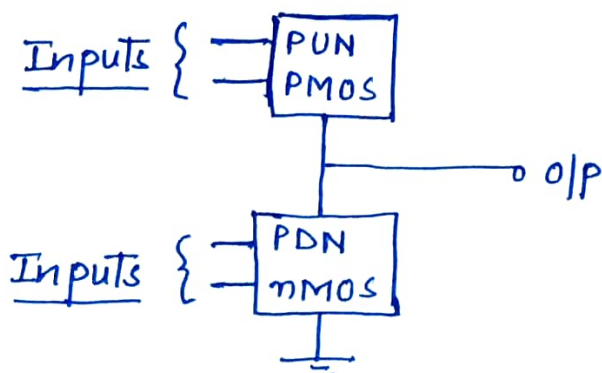
CMOS AS NOR Gate :- NOR Gate:- $A \rightarrow$  $y = \overline{A+B}$

Truth Table of NOR Gate:-

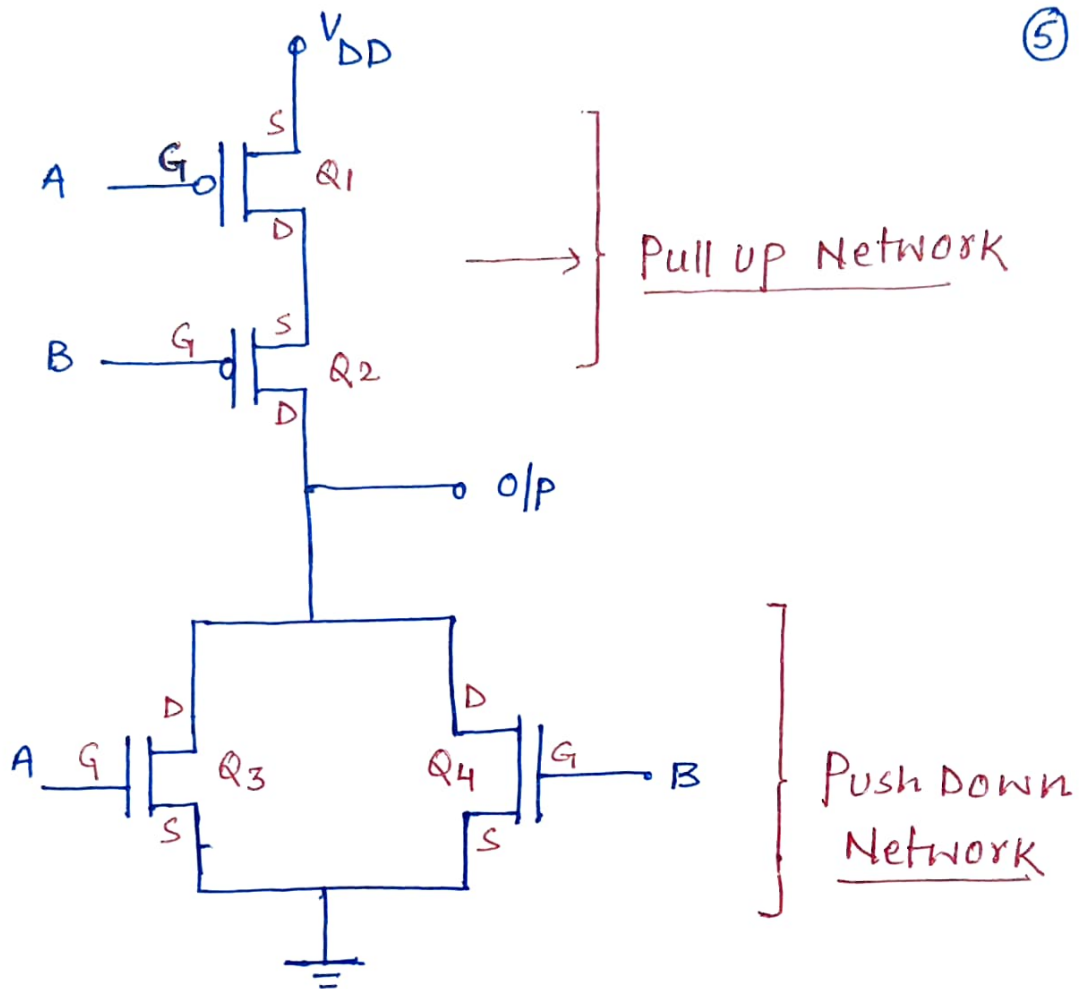
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

⇒ In CMOS AS NOR Gate, We perform (+) operation.

⇒ $\left. \begin{array}{l} \text{NMOS} \rightarrow \text{Connected in parallel} \\ \text{PMOS} \rightarrow \text{Connected in series} \end{array} \right\}$



⑤



Truth Table :-

Inputs		PMOS		NMOS		Output
		Q_1	Q_2	Q_3	Q_4	
A	B					
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0